# $\gamma_{\text{PrEP}}$ Payer-Mix Sensitivity Analysis:
## Meeting Follow-Up and Revised-Parameter Results

August 13, 2026

**Abstract**

Part I of this memo reproduces the August 10, 2026 meeting follow-up (`INFORM-HIV-Analysis/Raff_tmp/gamma_prep_meeting_followup.tex`), which identified that the model's PrEP private-insurance-floor parameter ($r = 0.78$, national) is likely too high for INFORM-HIV's population (young Black MSM/TGW in Chicago) and proposed a working value of $r \approx 25\%$. Part II presents the results of actually running that revised value through the cost-mapping pipeline: a point-estimate comparison against the current committed baseline, and a small two-parameter exploratory sweep over $r$ and the baseline PrEP coverage anchor $P^0$.

# Contents

# Part I
# Meeting Follow-Up (2026-08-10)

## 1 Jonathan's question: why is it $P - \gamma$, not $P \times \gamma$?

The coverage model (main text, Eqs. 1–2) is:

$$P_{\text{ART}} = \min\left(1,\ \frac{G_{\text{ART}}}{\alpha\, C_{\text{ART}}\, N_{\text{HIV+}}} + \gamma_{\text{ART}}\right), \tag{1}$$

$$P_{\text{PrEP}} = \min\left(1,\ \frac{G_{\text{PrEP}}}{\beta\, C_{\text{PrEP}}\, N_{\text{HIV-}}} + \gamma_{\text{PrEP}}\right). \tag{2}$$

Jonathan's question concerned the calibration equations (supplement, Eqs. 4–5), which solve for $\alpha$ and $\beta$ given an observed baseline coverage $P^0$:

$$\alpha = \frac{G_{\text{ART}}}{(P^0_{\text{ART}} - \gamma_{\text{ART}})\, C_{\text{ART}}\, N_{\text{HIV+}}}, \tag{3}$$

$$\beta = \frac{G_{\text{PrEP}}}{(P^0_{\text{PrEP}} - \gamma_{\text{PrEP}})\, C_{\text{PrEP}}\, N_{\text{HIV-}}}. \tag{4}$$

Why $(P^0 - \gamma)$ and not $(P^0 \times \gamma)$?

### 1.1 The subtraction is algebra, not a separate modeling choice

Equations (3)–(4) are not an independent assumption. They are what you get by taking the *forward* model, Eqs. (1)–(2), and solving for $\alpha$ (or $\beta$) at the observed baseline. Concretely, for ART:

$$\begin{aligned}
P^0_{\text{ART}} &= \frac{G_{\text{ART}}}{\alpha\, C_{\text{ART}}\, N_{\text{HIV+}}} + \gamma_{\text{ART}} \\
\implies \quad \underbrace{P^0_{\text{ART}} - \gamma_{\text{ART}}}_{\text{government-attributable share}} &= \frac{G_{\text{ART}}}{\alpha\, C_{\text{ART}}\, N_{\text{HIV+}}} \\
\implies \quad \alpha &= \frac{G_{\text{ART}}}{(P^0_{\text{ART}} - \gamma_{\text{ART}})\, C_{\text{ART}}\, N_{\text{HIV+}}}.
\end{aligned}$$

The subtraction appears because the forward model is **additive**: total coverage is a private floor $\gamma$ (assumed fixed, unaffected by government spending) *plus* a government-driven increment $G/(\alpha CN)$, capped at 1. Isolating "how much of the observed coverage is attributable to government spending" is therefore just $P^0$ minus the part we've already assumed is privately financed — a subtraction, by construction.

## 1.2 The modeling choice that actually matters

The real assumption is one level up: coverage is additive (private floor + government increment), not multiplicative (e.g., government funding scaling up access among the population *not* already privately covered, or some other interaction form). We chose the additive form for parsimony; its limitations (nonlinearity, insurance-stratum heterogeneity, threshold effects) are flagged in the main text as directions for future refinement.

## 1.3 Why this means $\alpha, \beta$ are not free parameters

A direct consequence, and the reason this matters for the meeting's discussion: once $\gamma$ and the observed baseline $P^0$ are fixed, $\alpha$ (or $\beta$) is *determined* by Eq. (3)/(4) — it is not independently chosen or estimated. This means we **cannot revise $\gamma_{\mathbf{PrEP}}$ in isolation**: any change to $\gamma_{\text{PrEP}}$ requires recalibrating $\beta$ in the same step, or the model will no longer reproduce the observed baseline PrEP coverage.

## 1.4 A useful analytical fact: what actually drives cut-sensitivity

It's worth separating two questions:

- **Question 1 (coverage level):** what is $P^0_{\text{PrEP,obs}}$ for young Black MSM/TGW in Chicago? (Current model: 36%, national. Evidence suggests 20–24% may be more appropriate for this population.)
- **Question 2 (payer mix):** what fraction $r$ of PrEP users in this population have private/non-public coverage that would survive a government funding cut? (Current model: $r = 0.78$, national, giving $\gamma_{\text{PrEP}} = r \times P^0 = 0.78 \times 0.36 \approx 0.28$.)

Define the *pass-through fraction* — the share of observed baseline coverage that would be lost under a 100% funding cut (i.e., $\Delta = 1$, so $P \to \gamma$):

$$\text{pass-through} = \frac{P^0 - \gamma}{P^0} = \frac{P^0 - rP^0}{P^0} = 1 - r.$$

For PrEP, $1 - 0.78 = 22\%$; for ART, $1 - \gamma_{\text{ART}}/P^0_{\text{ART}} = 1 - 0.15/0.62 \approx 76\%$.

*The important implication: the* relative *pass-through fraction depends only on $r$ (Question 2), not on $P^0$ (Question 1), because $\gamma = rP^0$ by construction, so $P^0$ cancels out of the ratio. If we only revise the coverage-level anchor (36% → 24%) while keeping the national $r = 0.78$, the model's relative sensitivity to funding cuts is* unchanged *(still ∼22% pass-through). Only the absolute number of percentage points at risk would shrink. Part II of this memo verifies this claim numerically.*

# 2 Summary of the 2026-08-10 meeting

## 2.1 The problem, restated

- $\gamma_{\text{PrEP}} = 0.28$ was derived entirely from national data: a national baseline PrEP coverage of 36% (CDC Monitoring 2022) and a national payer-mix estimate that 78% of PrEP users are privately insured (CDC NHBS 2024 / Bonacci et al. 2023).
- INFORM-HIV simulates a specific population — young Black MSM/TGW in Chicago — for whom both numbers are likely wrong in the same direction: NHBS puts Black MSM PrEP use at ∼24% nationally (vs. 57% for White MSM), and Black MSM are known to rely far more on Medicaid/Ryan White/ADAP and less on private insurance than PrEP users nationally.
- If $r = 0.78$ is too high for this population, the model currently overstates how much PrEP access survives a funding cut, and therefore *understates* the harm of PrEP funding cuts specifically for the population the paper is about.

## 2.2 What we checked, and what we found

- **John Peller's (AFC) email**, forwarded by Anna, estimates 32–33% of Illinois residents are covered by ERISA-governed self-insured employer plans, 23–25% by state-regulated fully insured plans, 35–36% by public programs, and 6–7% are uninsured. This addresses a different policy channel (removal of the USPSTF Grade A PrEP mandate) rather than the government-funding-reduction channel our model represents, but is a useful sanity-check bound: combined commercial coverage state-wide is only ∼55–58%, well below the 78% assumed among PrEP users — consistent with, but not proof of, $r = 0.78$ being too high for a lower-income, more publicly-insured subpopulation.
- A targeted literature search (AIDSVu/PrEPVu.org, CDC MMWR 2019 23-urban-areas NHBS report, the AJPH 2021 Medicaid-expansion NHBS study covering 22 cities including Chicago, and a 2018–2022 pharmacoequity claims analysis) found that **no source was identified in our targeted search that crosses race/ethnicity with PrEP-payer type**. This looks like a genuine gap in the literature, though our search was not exhaustive and is not documented here with full search terms, dates, or a formal bibliography — treat this as a preliminary finding, not a systematic review.
- The Black–white PrEP-use gap persists *even among the insured*, and the AJPH study found the racial gap persists regardless of Medicaid-expansion status — insurance/payer source is not the whole story; a revised $r$ corrects the financing piece but should not be read as fully correcting for the underlying access disparity.
- Meg McElroy's ADAP FOIA summary (Illinois-specific, uninsured vs. insured client counts, April 2024–March 2025) could serve as a partial cross-check, though it is not stratified by race.

### 2.3 Where the discussion landed

We agreed to set $r \approx 25\%$ as a working value:

- this is a **post-hoc recalculation** — it only touches the funding-to-coverage layer, not the underlying ABM, so no new simulation is required;
- $\beta$ must be recalibrated alongside any revision to $\gamma_{\text{PrEP}}$ (Section 1);
- getting an expert read from AFC (John Peller/Meg McElroy) on a defensible number would be more credible than an internal guess.

## 3 Action plan (from the 2026-08-10 meeting)

1. Settle on a working value/range for $r$ (equivalently, $\gamma_{\text{PrEP}}$).
2. Recalibrate $\beta$ (and audit $\gamma_{\text{ART}}/\alpha$ the same way) using Eqs. (3)–(4).
3. Send AFC a narrower, specific data request: among PrEP users who are young Black MSM/TGW in Chicago, what fraction access PrEP via private insurance vs. Medicaid/ADAP/PrEP4Illinois/uninsured?
4. **Re-run the cost-mapping layer with the revised parameters** — regenerating the forest plot, the ribbon+heatmap panel, and the scenario table — and mirror the same parameter change in `INFORM-HIV-App`.
5. Document the limitation explicitly in the supplementary appendix: a race/city-stratified PrEP payer-mix estimate does not appear to exist in the published literature.

Part II below carries out item 4 above as an exploratory sensitivity check (not yet the paper's committed figures — see Section 7.1).

# Part II
# Sensitivity Analysis Results (2026-08-13)

## 4 What was run

Using `04_cost_mapping/sensitivity/` (branch `sensitivity/gamma-prep-r-p0` of `INFORM-HIV-Funding-Scenario-Exploration`), we recalibrated $\gamma_{\text{PrEP}}$ and $\beta_{\text{PrEP}}$ from a chosen $(r, P^0)$ pair via Eqs. (3)–(4) (`R/recalibrate.R`), then queried the same stage-03 incidence surrogate and common-random-numbers machinery (`compute_scenario_draws_crn()`) the main-text forest plot uses, without rebuilding the full $101 \times 101$ grid. Three parameter sets:

- **Baseline** — current committed values: $r = 0.78$, $P^0 \approx 36\%$ (i.e. $\gamma_{\text{PrEP}} = 0.28$, $\beta_{\text{PrEP}} = 4.0$).
- **Case 1** — $r = 25\%$, $P^0$ unchanged ($\approx 36\%$).
- **Case 2** — $r = 25\%$, $P^0 = 24\%$ (both the payer-mix and the baseline-coverage-level anchor revised).

Two important scope notes: (1) this uses only the pure-computation modules (`parameters.R`, `irr_common_random_numbers.R`) and plain `ggplot2` output — it does **not** use `plot_forest.R`/`plot_ribbon.R`, which require the RAND house-style package `randplot` that is not currently installed and will need to be requested from Pedro before the paper's actual figures can be regenerated with the revised parameter. (2) Nothing in `cost_params.yml` or the tracked `output/` directory on `main` was touched — the paper's currently committed $r = 0.78$ baseline is unchanged.

# 5 Case comparison: current vs. revised $r$

Table 1 reports the mean incidence rate ratio (IRR, averaged over the 0–10 year trajectory horizon, vs. the no-cut baseline) with 95% credible intervals, for three government-funding-reduction scenarios at 10/25/40% cuts. **Case 1 and Case 2 are numerically identical** on every row (verified to floating-point precision by construction, not just approximately) — directly confirming the analytical claim in Section 1: $(P^0 - \gamma_{\text{PrEP}})/P^0 = 1 - r$ is algebraically independent of $P^0$. Only $r$ is shown in the table below as a result.

*A note on what these intervals do and do not capture: they are the surrogate's posterior predictive intervals at a fixed $(r, P^0)$ pair, propagating only the fitted incidence surrogate's own predictive uncertainty. They do not propagate uncertainty in $r$ or $P^0$ themselves, in costs or funding totals, or in the additive coverage-model structure (Section 1) — those are separate, unquantified sources of uncertainty layered on top. Section 6 explores sensitivity to $r$ and $P^0$ directly, but as a parameter sweep, not as an interval that combines with the credible intervals below.*

The ART-only rows are identical between columns by construction (the ART payer-mix parameter $\gamma_{\text{ART}}$ was not touched); the PrEP-only and combined rows show a consistently larger incidence-rate-ratio under the revised, lower $r$ — the effect widens as the funding cut deepens.

# 6 Two-parameter exploratory sweep around $r = 25\%$

This is a parameter sweep, not an uncertainty quantification: it explores how the headline result moves *locally*, in a band around the proposed working value, not a robustness check against the full plausible range between the current committed $r = 0.78$ and the proposal. We drew $N = 200$ independent joint samples of $r \sim \text{Uniform}(0.125, 0.375)$ and $P^0 \sim \text{Uniform}(0.12, 0.36)$ (each $\pm 50\%$ around the Case 2 center value, $r = 25\%/P^0 = 24\%$ — a uniform distribution has no mode, so "center" rather than "mode" is the accurate

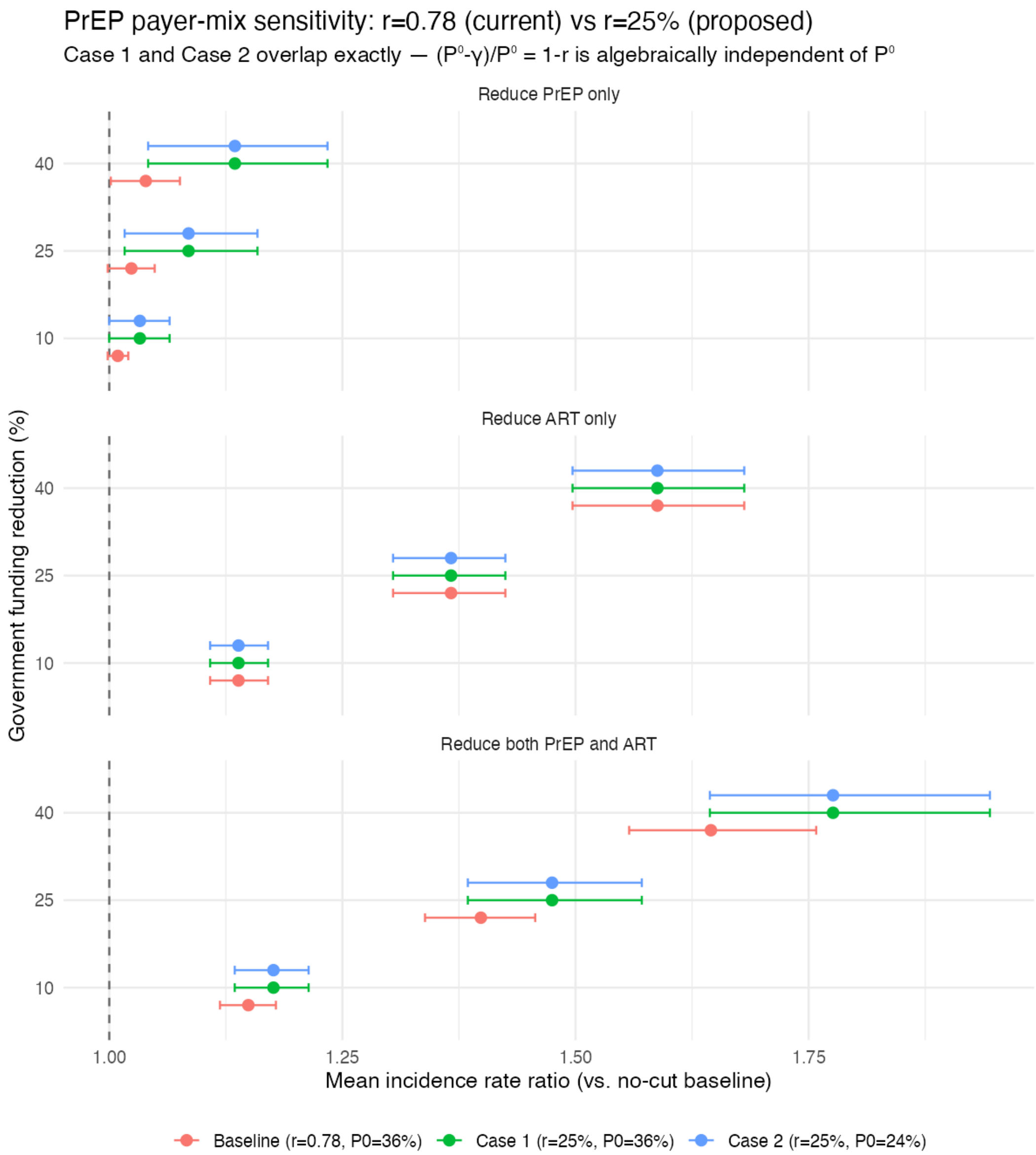


Figure 1: Forest-style comparison of mean incidence rate ratio across the three funding-cut scenarios, current ($r = 0.78$) vs. revised ($r = 25\%$, both $P^0$ variants). Case 1 and Case 2 overlap exactly.

Table 1: Mean incidence rate ratio (95% CI) by scenario and funding-reduction level.

| Scenario | Funding cut | Current ($r = 0.78$) | Revised ($r = 25\%$) |
|---|---|---|---|
| Reduce PrEP only | 10% | 1.009 [0.998, 1.020] | 1.033 [1.000, 1.065] |
| | 25% | 1.024 [0.998, 1.049] | 1.085 [1.016, 1.159] |
| | 40% | 1.039 [1.002, 1.076] | 1.135 [1.042, 1.234] |
| Reduce ART only | 10% | 1.139 [1.108, 1.170] | 1.139 [1.108, 1.170] |
| | 25% | 1.367 [1.304, 1.425] | 1.367 [1.304, 1.425] |
| | 40% | 1.588 [1.497, 1.681] | 1.588 [1.497, 1.681] |
| Reduce both PrEP and ART | 10% | 1.149 [1.119, 1.179] | 1.176 [1.135, 1.214] |
| | 25% | 1.399 [1.339, 1.457] | 1.475 [1.385, 1.571] |
| | 40% | 1.645 [1.558, 1.758] | 1.776 [1.644, 1.944] |

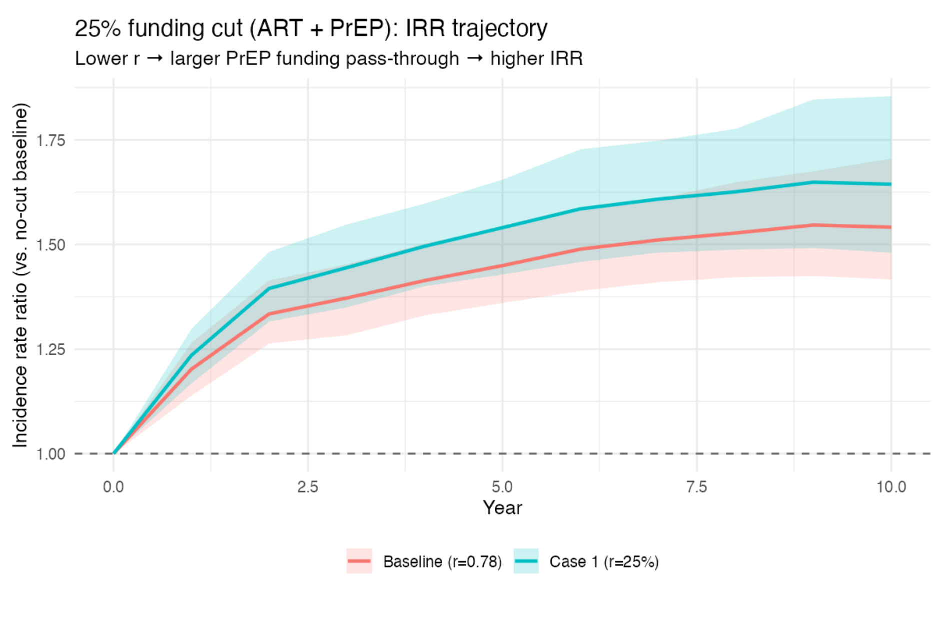


Figure 2: Year-by-year incidence-rate-ratio trajectory for a 25% government funding cut applied to both ART and PrEP simultaneously, current vs. revised $r$.

term), recalibrated $\gamma_{\mathrm{PrEP}}/\beta_{\mathrm{PrEP}}$ for each draw, and queried the single headline scenario (25% government funding cut, both ART and PrEP) for each. *Because the sampled $r$ range (0.125–0.375) sits entirely below the current $r = 0.78$, every draw necessarily shows more harm than the current baseline by construction of the range — that is not, on its own, independent evidence that the harm-increase conclusion is robust across the full space from current to proposed. What this sweep* does *legitimately establish is the first two bullets below: within that local band, the relationship is (a) close to linear in $r$ and (b) flat in $P^0$, which is the direct empirical trace of the algebraic relationship in Section 1, not new evidence about the epidemiology.*

- IRR ranged from 1.457 to 1.491 across the 200 draws (mean 1.474).
- $\mathrm{cor}(\mathrm{IRR}, r) = -1.000$ — an almost perfectly linear relationship *within this local band*; lower $r$ (more Medicaid/ADAP-reliant payer mix) means a larger funding pass-through and a higher incidence rate ratio.
- $\mathrm{cor}(\mathrm{IRR}, P^0) = -0.016$ — statistically and practically zero, the empirical confirmation of the algebraic claim that $P^0$ does not affect the incidence/IRR outcomes *within the current post-hoc funding-to-coverage mapping* (see the Discussion below for what this does and does not imply more broadly).

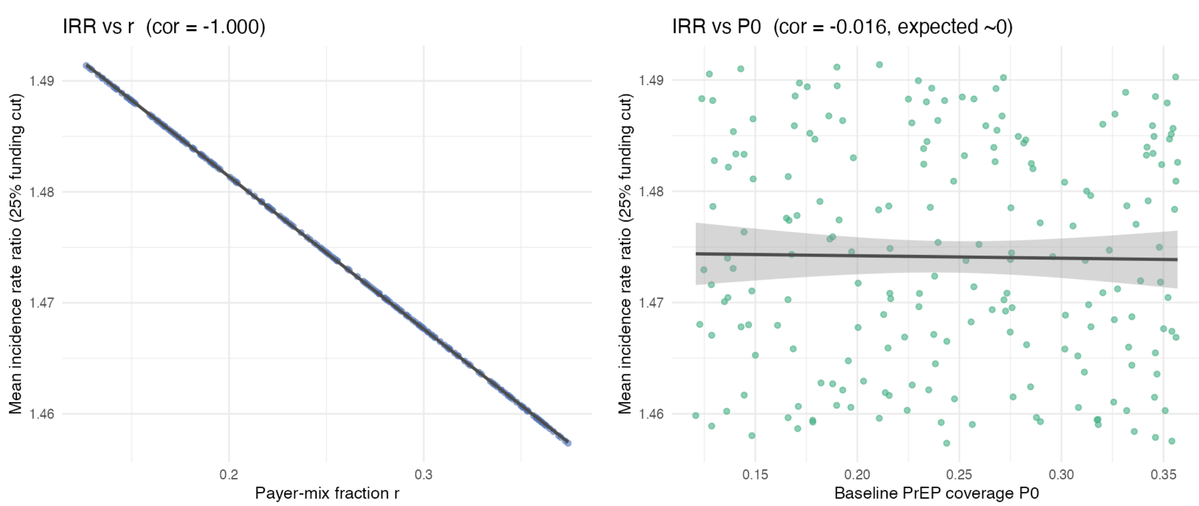


Figure 3: Parameter sweep ($N = 200$ draws): incidence rate ratio at a 25% funding cut vs. $r$ (left, strong negative relationship) and vs. $P^0$ (right, no relationship).

# 7 Discussion

The revised, *provisional* payer-mix assumption ($r \approx 25\%$ rather than the national $r = 0.78$) produces a materially larger projected harm from PrEP funding cuts than the paper's currently committed baseline, consistent with the meeting's hypothesis that the national figure

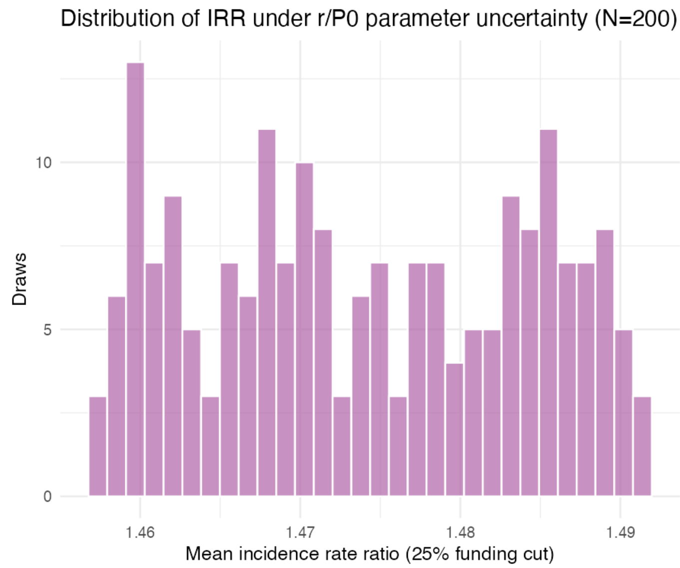


Figure 4: Distribution of the 25%-funding-cut incidence rate ratio across the 200 $(r, P^0)$ draws.

understates this population's exposure. *We deliberately avoid calling $r = 25\%$ "population-appropriate" here: as Section 2 already says, no source crossing race/ethnicity with PrEP-payer type was found, so this is a working/illustrative scenario value from expert elicitation, not an empirically estimated parameter. There is also a definitional looseness worth flagging: $r$ is operationalized as "fraction privately insured," used as a proxy for "fraction whose PrEP access is immune to* this specific *government funding cut." Those are not the same thing — private insurance does not guarantee complete immunity to a funding cut, and some public-financing categories (e.g. state programs funded independently of the federal lines being cut) may also be unaffected by it. Sharpening $r$'s operational definition is a natural companion task to getting AFC's number.* Within the locally-sampled band around this proposed value (Section 6), the sign and rough magnitude of the shift do not change — but that band was centered on the proposal and does not extend up to the current $r = 0.78$, so this is evidence of local stability near the proposed value, not a robustness claim spanning the full current-to-proposed range.

Separately, the analysis is — as anticipated analytically — insensitive to the baseline PrEP coverage-level anchor $P^0$, but only *within the current post-hoc funding-to-coverage mapping layered on top of the existing, already-fitted ABM/surrogate*: $P^0$ cancels out of the coverage-reduction fraction fed to the surrogate query, by construction of $\gamma_{\text{PrEP}} = rP^0$. *This does not mean baseline PrEP coverage is epidemiologically irrelevant in general, or that a population-specific $P^0 = 24\%$ has been validated as a substitute for the $\approx 36\%$ the ABM/surrogate was actually built around. If $P^0 = 24\%$ is intended as a genuinely corrected baseline for the simulated population — not just a bookkeeping input to this cost-mapping layer — that would require assessing whether the underlying ABM calibration and surrogate need to be revisited, which is outside the scope of this post-hoc analysis.* This means the single highest-value next step from the memo's action plan (Section 3, item 3) is getting AFC's read on a defensible $r$; refining $P^0$ alone, within the current architecture, would not change the funding-cut conclusions reported here.

### 7.1 Status

This is an exploratory sensitivity check, not a revision to the paper. The committed baseline on `main` ($r = 0.78$) is unchanged. Before the revised parameter can be adopted in the paper's actual figures (which use the RAND house style via `randplot`), we need: (1) AFC's input on a defensible $r$, and (2) the `randplot` package, which is not currently installed anywhere accessible and will be requested from Pedro. Item 4 of the action plan (mirroring the change into `INFORM-HIV-App`) is likewise deferred until $r$ is actually adopted.